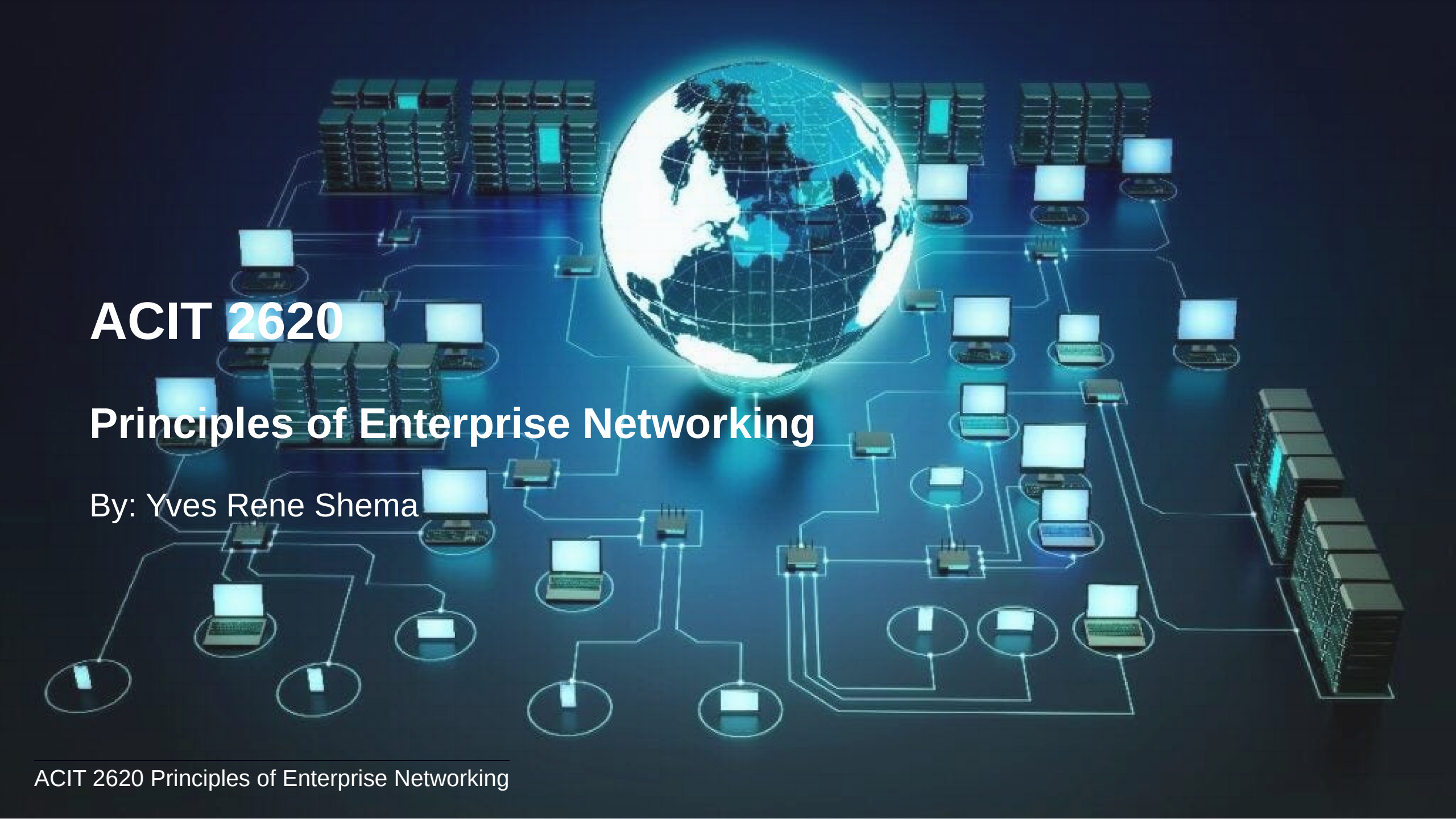


ACIT 2620

Principles of Enterprise Networking

By: Yves Rene Shema

What is a network?

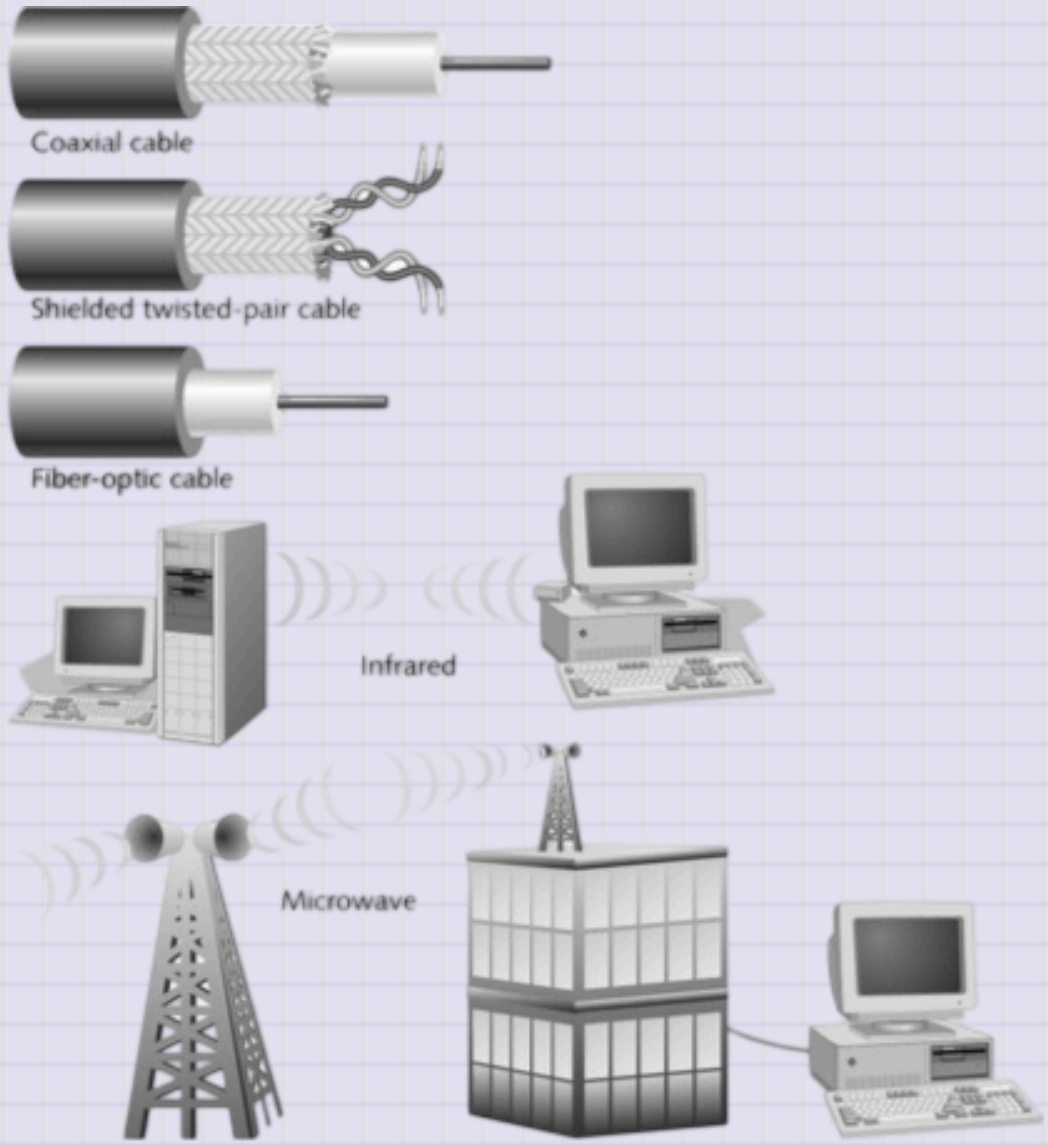
Network

- Collection of `nodes` connected by some type of transmission media or `link`, for the purpose of sharing services, devices or data (i.e. networked `resources`)

Node

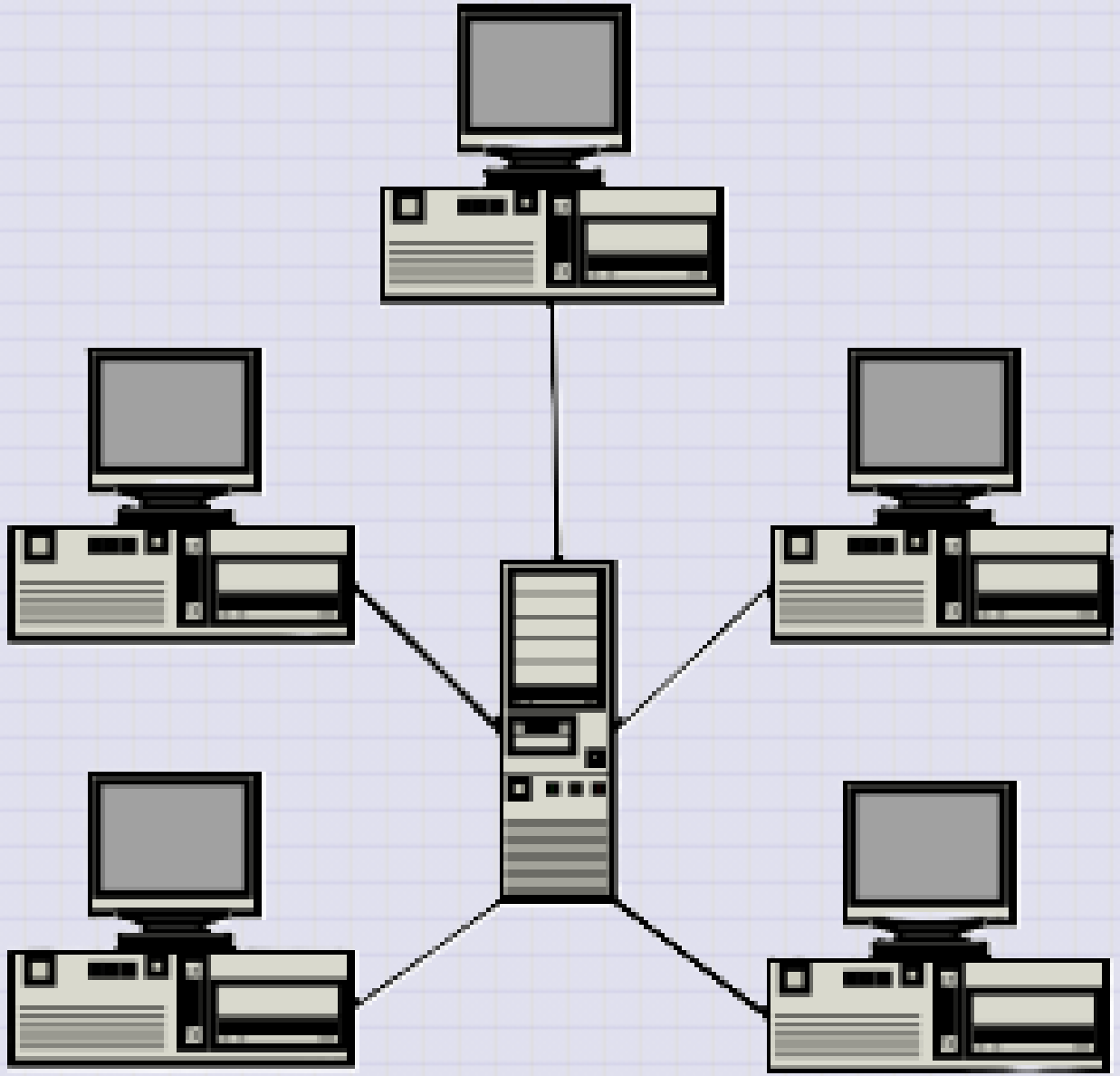
- Any device that can communicate over the network and is identified by a unique identifying number, known as its `network address`.

Link

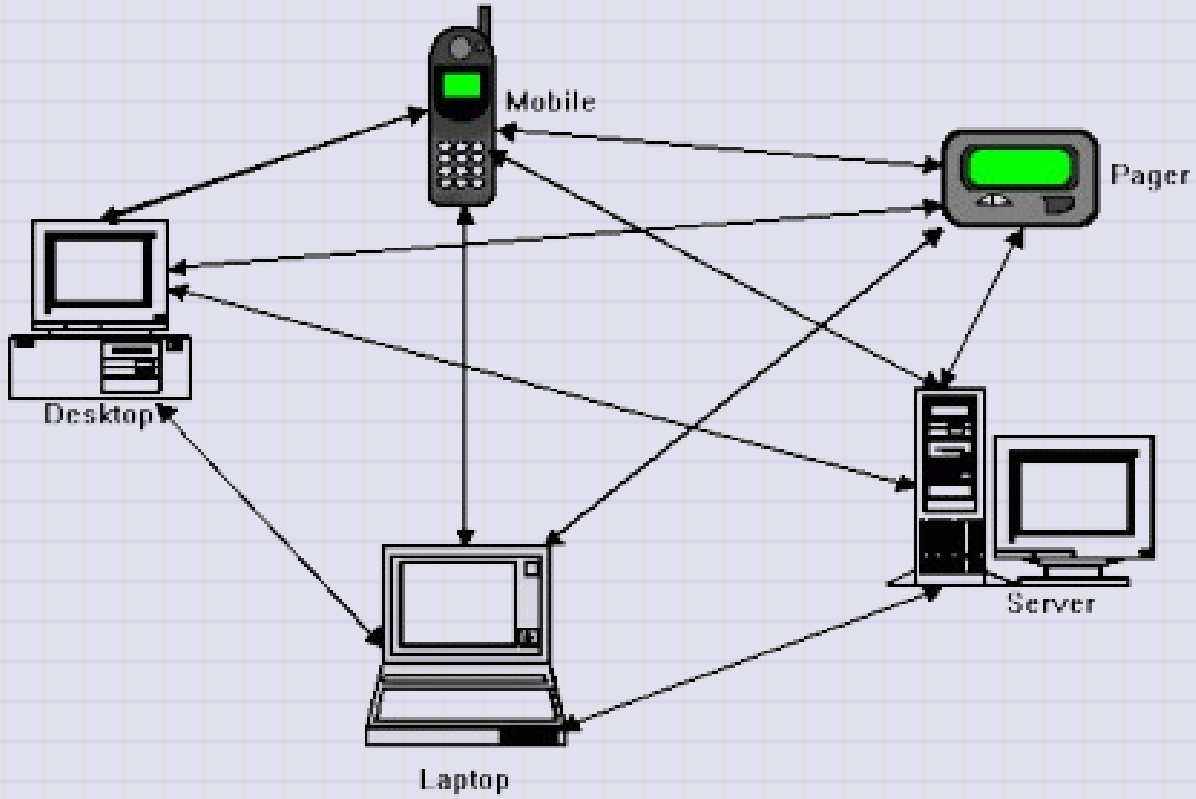

Coaxial cable
Shielded twisted-pair cable
Fiber-optic cable
Infrared
Microwave

Resource Control

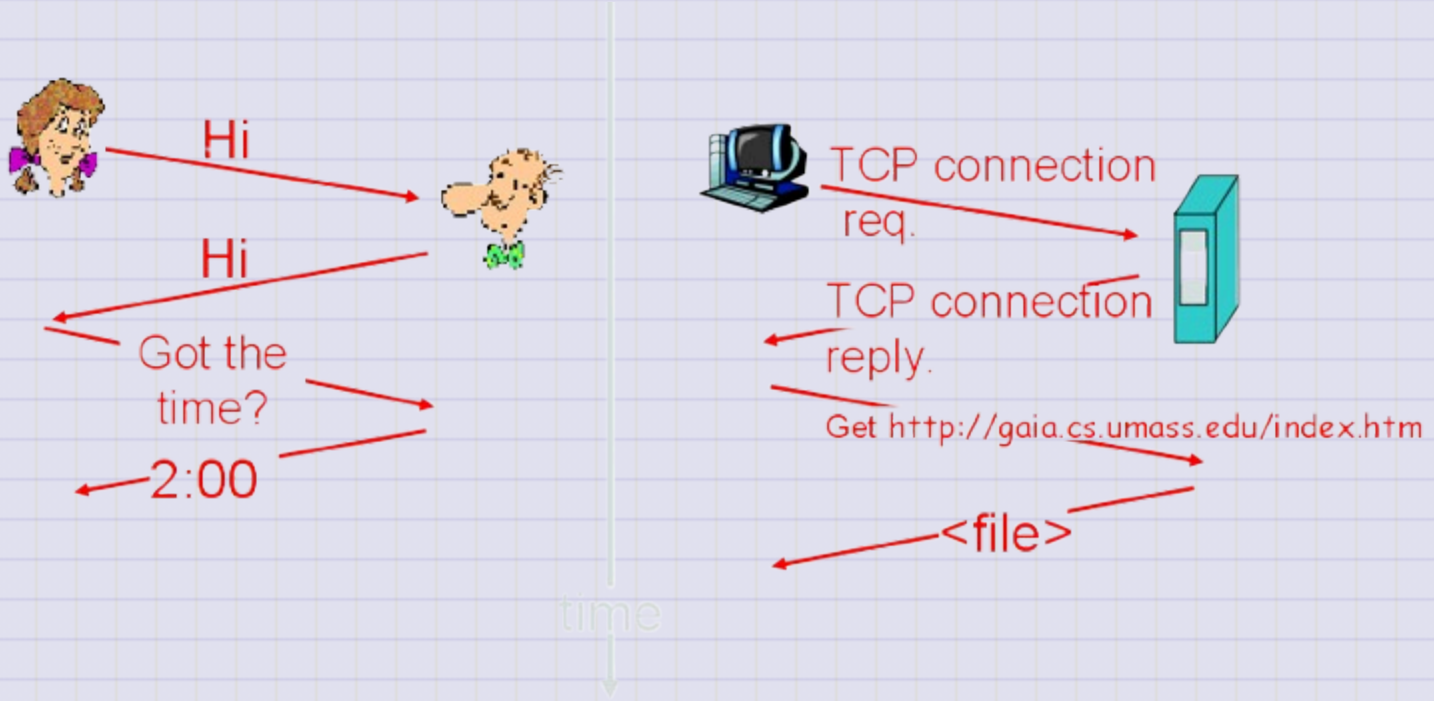
Client-server Networks



Peer-to-peer Networks



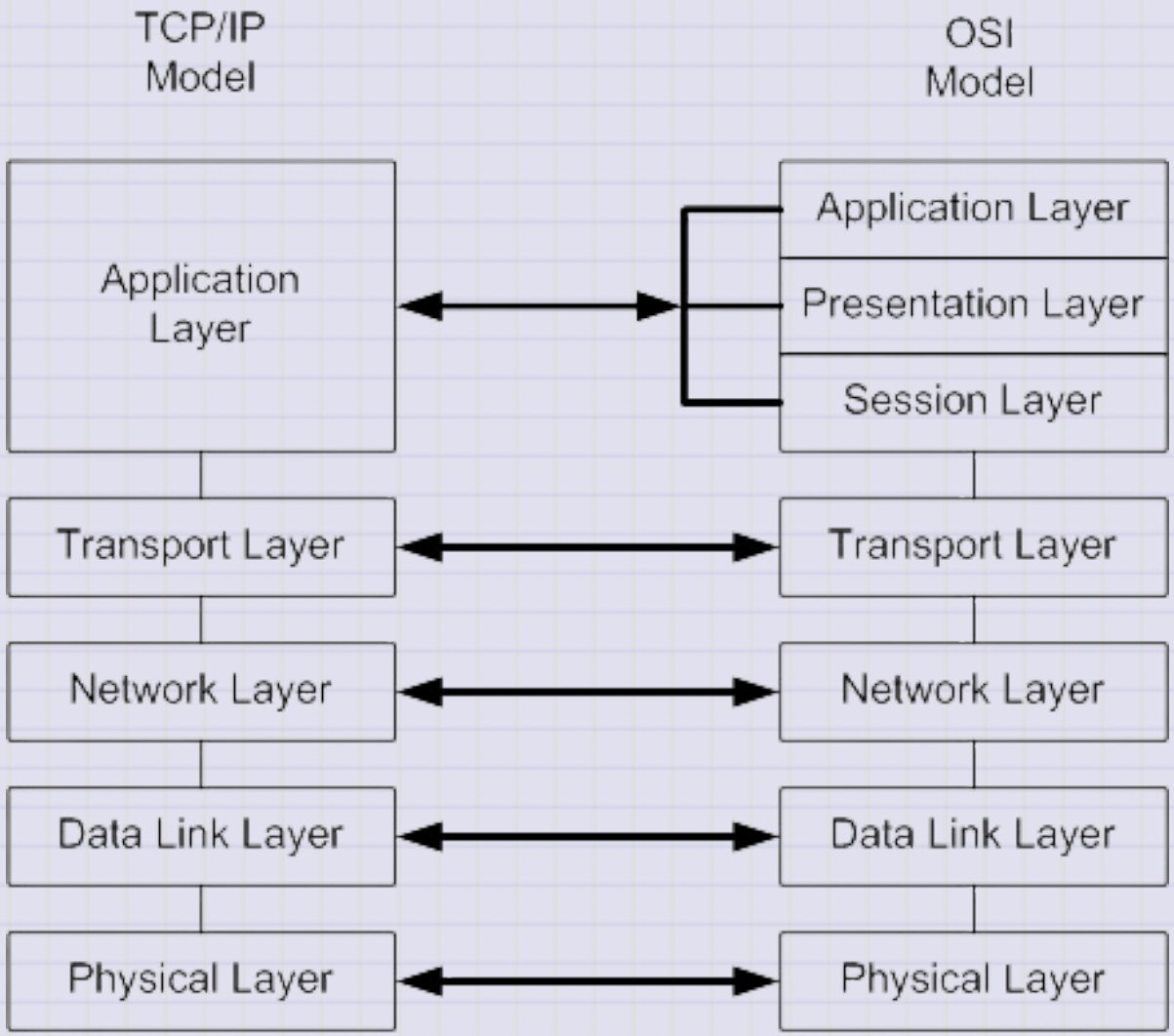
Protocols


Hi
Hi
Got the time?
2:00
TCP connection req.
TCP connection reply.
Get http://gaia.cs.umass.edu/index.htm
<file>
time

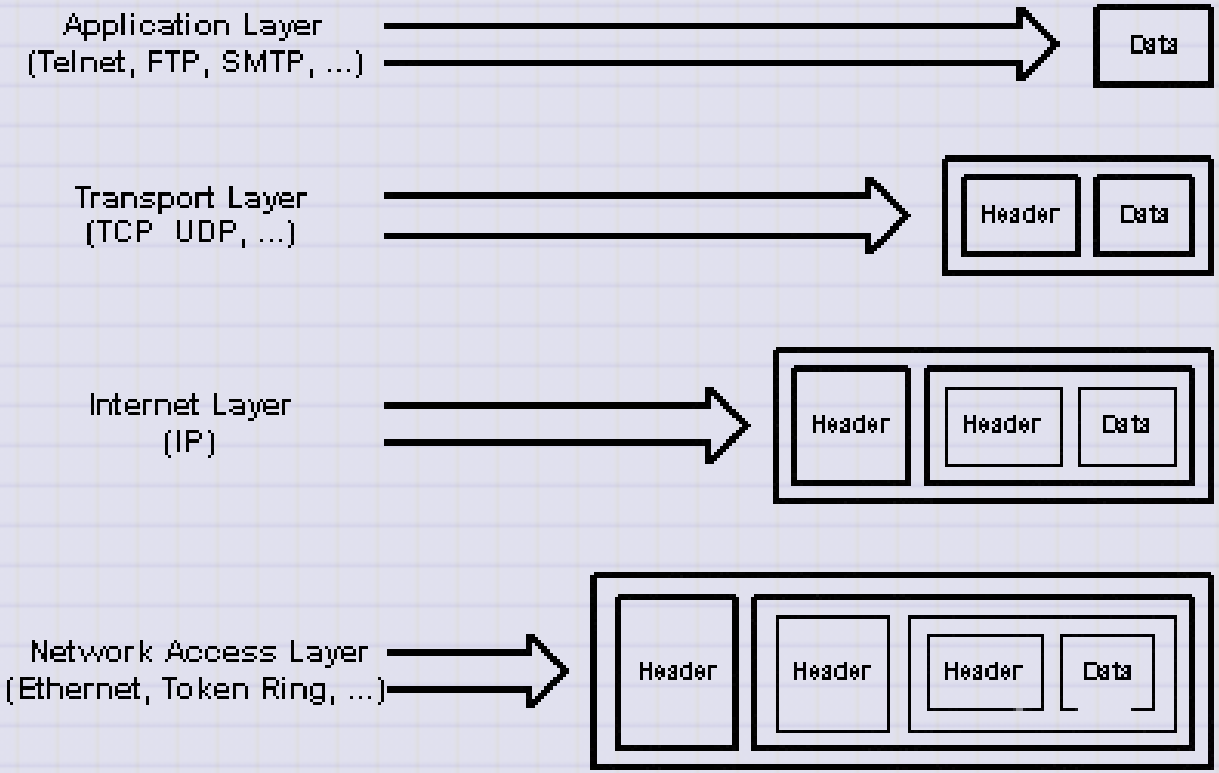
- Mutually agreed upon rules for communication
- Define the format, order of messages sent and received among network entities, and actions taken upon transmission, receipt, and timeout
- Govern all communication activity on the internet

Example Protocols

Layered networking model



Encapsulation



Protocol Data Units (PDU)

